

112A Introduction to Computer and Computer Science

Homework Assignment #8

Due: 12/8 12:00:00

In this homework assignment, you will practice how to apply the concept of object-oriented programming to solve a practical issue. As you have learned all the related knowledge during the lectures, you should have the ability to accomplish this.

Problem #1: A Dice Game (50%)

You have already created a “**Dice**” and “**Person**” object in the [homework#7](#). This time, we are going to use those two objects to create a **game**. Here are the rules of this **game**:

1. At the beginning of this game, 2 players have the same amount of dices. The dice type of these 2 players are in the same.
2. There is a banned list with at least one integer in the range of $[1, D_{max}]$, where D_{max} represents the maximum outcome of dice.
3. Each round, every player will roll their dices and then remove the corresponding dices whose point is equal to the element in the banned list.
4. If the dice box of the player is empty, she/he is eliminated from this game.
5. Repeat step 3 and 4 until only one person left.

Here is the sample code:

```
def oneGame(player1, player2, numDices=10, diceType=12, ban=None):
    p1 = Person(player1, numDices=numDices, diceType=diceType)
    p2 = Person(player2, numDices=numDices, diceType=diceType)

    ???
```

If everything goes well, you might have this result:

```
(nycudopcs) 0:\                                \Homeworks\112A\HW8\112A_hw8_0123456789>python main_hw8.py --name1
"Chung Bo-Eat" --name2 "Chen Yo-Thai" --num 5 --dtype 4 --ban 1 2 3 --plot True --games 100
Problem 1:
Round 1:
    Current state of Chung Bo-Eat: [1, 2, 2, 4, 1]
    Current state of Chen Yo-Thai: [2, 2, 1, 2, 1]
===== Removing dices =====
    Current state of Chung Bo-Eat: [4]
    Current state of Chen Yo-Thai: []
End of round 1.
=====
Chung Bo-Eat wins!
```

Problem #2: Statistics of N Games (50%)

So far, we have created enough functions and objects to do the next experiment, the statistics of N games. Please write a function called “**gameStats**” which aims to return the corresponding result of N games. For example, it should show this message:

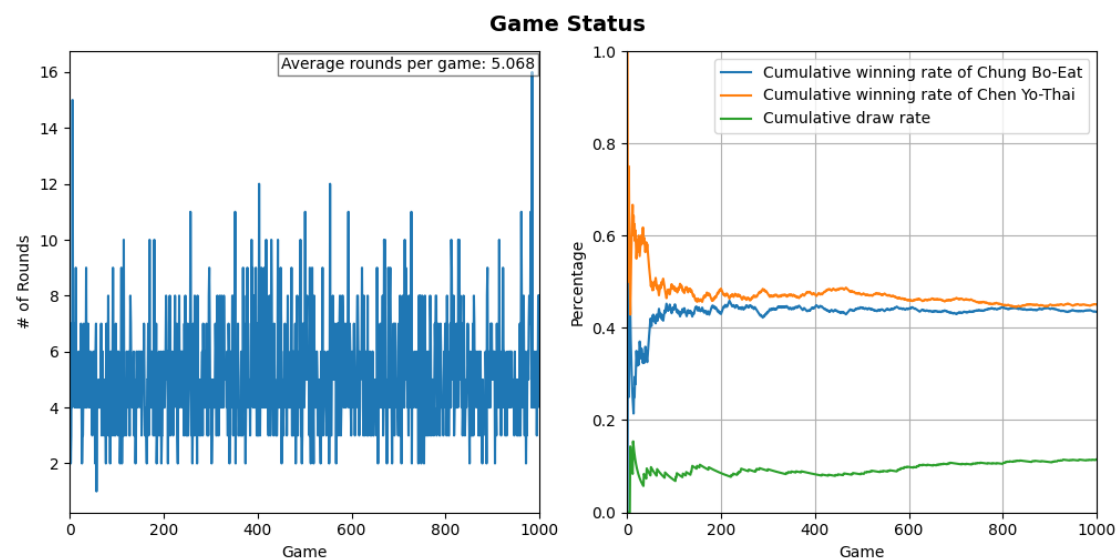
```
(nycudopcs) 0:\Homeworks\112A\HW8\112A_hw8_0123456789>python main_hw8.py --name1
"Chung Bo-Eat" --name2 "Chen Yo-Thai" --num 6 --dtype 12 --games 100
Problem 2
Stats of 100 games:
  Chung Bo-Eat wins: 37
  Chen Yo-Thai wins: 48
  Draw: 15
```

(Bonus) Problem #3: Plot the result of N Games (50%)

Please plot the result of the statistics of N games.

```
(nycudopcs) 0:\Homeworks\112A\HW8\112A_hw8_0123456789>python main_hw8.py
--name1 "Chung Bo-Eat" --name2 "Chen Yo-Thai" --plot True
Problem 3
Stats of 1000 games:
  Chung Bo-Eat wins: 435
  Chen Yo-Thai wins: 450
  Draw: 115
```

Here is a template. You only need to provide the same information. Please feel free to change the style of this figure.

**FRIENDLY REMINDER**

There must be a method attribute in your “**Person**” to “**remove**” the dice.
In this homework, you can use **NumPy**, **Matplotlib**, and **random** to help you.

Please accomplish this homework with an organized code (e.g., with main script and function script). **Your main script file name must be 'main_hw8.py.'** Here is a template for your code structure:

```
112A_hw8_0123456789
├─ obj.py          # Module 1 for hw8
├─ func.py         # Module 2 for hw8
├─ :              # Other modules if necessary
└─ main_hw8.py     # Main script of hw8
```

FRIENDLY REMINDER

You don't need to follow this structure exactly, just keep your main script clean.

Here is a template for your main script.

```
import argparse
from func import oneGame, gameStats

if __name__ == "__main__":
    parser = argparse.ArgumentParser(description="2023 NYCUDOPCS
Homework #8")
    parser.add_argument("--num", default=10, type=int, help="Number of
dices (default: 10)")
    parser.add_argument("--dtype", default=12, type=int, help="Dice type
(default: 12)")
    ???

    opt = parser.parse_args()

    print("Problem 1:")
    oneGame(
        player1=opt.name1,
        player2=opt.name2,
        numDices=opt.num,
        diceType=opt.dtype,
        ban=opt.ban,
    )
```

```
print("="*60)

print("Problem 2 & 3")

nRounds, nGameStats, gameLog = gameStats(
    numGames=opt.games,
    player1=opt.name1,
    player2=opt.name2,
    numDices=opt.num,
    diceType=opt.dtype,
    ban=opt.ban,
    plot=opt.plot,
)
```

Hand in procedure:

As we had mentioned in the lecture, you should list all your collaborators in your programs. Here is the template:

```
""  
Created on Sun Aug 7 01:23:45 2023  
  
@author: Xi Winnie, student ID  
  
@collaborators: EE Tsai, her student ID  
""
```

Please save your homework as a “.zip”, “.7z”, or “.rar” file, where the file name should follow this format:

112A_hw8_**ID**.zip

For example,

112A_hw8_0123456789.zip

Please note. **We are not going to accept any homework file with wrong file name, wrong file format, or without signature in main script.** Please double check the content of your file.

Once you have accomplished your works, you can submit your homework to the “E3@NYCU” system. There will be a section for uploading your homework.